

Dawson Metzger-Fleetwood

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EDUCATION

University of Maryland

B.S., Computer Science with Machine Learning Specialization

B.S., Finance

- Deans List
- Taisto Saloma Scholarship

College Park, MD

December 2022

Cumulative GPA: 3.65

9/10 semesters

Fall 2017-18

SKILLS

Programming: Swift, SwiftUI, UIKit, RealityKit, React, TypeScript, JavaScript, HTML/CSS, PHP, Python, Matlab

Software: Xcode, Xcode Instruments, XCUITest, Three.js, VSCode, Cursor, Claude Code, Node, Docker, Github Actions, Jira, Lambda, DynamoDB, S3, CloudFront, Cognito, SES, Matlab, Tensorflow, PyTorch, HuggingFace, LangChain, Pandas, Farama Gymnasium/PettingZoo

Languages: German (proficient), American Sign Language (proficient), Cued English (fluent)

WORK EXPERIENCE

About Objects.

Reston, VA

Software Engineer

July 2023 - Present

- Worked extensively on a multi-platform (iPad, iPhone, and Vision Pro) event management platform for government clients, featuring real-time resource tracking and 3D visualizations with RealityKit. Featured in Apple's enterprise showcase.
- Scaled the platform from inception to being able to handle 10,000+ concurrent users, architected real time data sharing for collaboration across distributed command teams, and optimized rendering hierarchical data with a tree-based architecture that reduced latency by 80%.
- Scaled a CI/CD pipeline with GitHub Actions and automated TestFlight builds as the team quadrupled in size. Mentoring interns and junior engineers.
- Worked on several internal projects, including leading the overhaul of the company website with custom WordPress plugins and a PHP backend, building an MCP server to help PMs with Jira project administration, and building pipelines to automate billing processes.
- Led development of a 3D ML object recognition pipeline, enabling rotation inference from 200 images. Supported an industrial oil drill manufacturer in analyzing drill bit component history and repairs.
- Led About Object's AI blog, writing articles on autoencoders, reinforcement learning, GNNs, and transformers.

Johns Hopkins University

Silver Spring, MD

Guest Lecturer and Consultant

Spring 2024

- Presented to interpreters on statistical concepts to enhance research interpretation, covering topics from fundamentals to the central limit theorem and multiple linear regression.
- Led the development of employee facing applications designed to integrate internal systems and improve preparation efficiency, including creating a custom MCP server.

Visual Language Associates, Inc

Silver Spring, MD

Technical Co-Founder

September 2019 - July 2023

- Co-founder of an agency that provides signed language interpreting and cued language transliteration services, successfully securing contracts with the MD and VA state governments, and serving over two dozen clients in the DMV area with an average contract value of over \$90,000.
- Led the automation and integration of client payment, payroll, and scheduling software to streamline operations.
- Responsible for coordinating scheduling with school officials and other interpreters and transliterators, as well as preparing and leading language and policy training sessions. Assisted with recruiting, onboarding, and sales.
- Developed methodology to screen employees, clients, and contracts, as well as match clients with service providers.
- Served as a cued language transliterator, directly providing interpretation/transliteration services to consumers.

TECHNICAL EXPERIENCE

Personal Projects

Swift ML Inference Library — Swift

Spring 2024

- Collaborated with a colleague to develop a Swift library for fast model training and inference, with the goal of showcasing Swift's potential as a standalone ML development language without reliance on C++.
- Designed the library to run entirely on the stack, avoiding performance degradations from heap usage.
- Utilized experimental Swift features such as non-copyable types and non-escapable closures, and created custom tensor types optimized for Apple's Unified Memory Architecture.

Protein Timing App — Swift

Spring 2022

- Developed an app using Swift and SwiftUI to help athletes time protein intake throughout the day.
- "Amino Amigo" is currently available on the iOS App Store, and has been downloaded in 11 countries including the UK, Australia, Brazil, Sweden, France, Angola, and New Zealand.

University of Maryland

Computer Vision — Matlab

Fall 2021

- Created four computer vision applications from scratch; object recognition and distance estimation software, a panorama stitching app, an Adobe rotobrush clone, and a drone location/flightpath tracker using raw camera data.

- Some of the techniques used included: GMMs, alternate color spaces (e.g. YCbCr), convolution, ANMS, RANSAC for homography estimation, cylindrical projection, bilinear interpolation for image blending, image segmentation using localized classifiers, motion estimation with color and shape models, SfM, SLAM, factor graphs, and GTSAM.

Computer Systems Architecture — C/C++ and Assembly

Fall 2022

- Developed an assembly language and error-checking software using only bit shifting, characters, and arrays in C.
- Implemented Tomasulo's algorithm, four branch predictors (G-share, Two-Level Adaptive, etc), and three cache coherency protocols (MOESI, etc) for parallelized processors (from 4 to 16 cores).
- Wrote Python scripts for extensive testing of different architectures/hyper-parameters.

ACTIVITIES/AFFILIATIONS

Startup Shell

College Park, MD

Fellow

September 2022 - Present

- Startup Shell is UMD's exclusive student-run startup incubator, founding home of ventures collectively worth over \$1 billion.
- As a fellow, was available to consult and assist with product development for a variety of ventures.
- Some ventures that launched during my tenure included a prompt engineering website with over 400,000 monthly visitors and a discord server with over 50,000 active members, a farmland valuation AI model, and a patient management platform for underfunded hospitals in Africa.

Boy Scouts of America

Colesville, MD

Eagle Scout

September 2012 — September 2016

- Led community service and leadership projects, including a weatherproof bulletin board construction for the troop's host church. Over 100 hours of service and multiple leadership roles (Patrol Leader, Assistant Senior Patrol Leader). Member of the Order of the Arrow